

The Comprehensive Academic and Research Dossier of Walter Rudametkin: Architecting Dynamic Resilience and Web Privacy

The professional identity of Walter Rudametkin represents a sophisticated convergence of software engineering rigor and empirical cybersecurity research. As a Full Professor at the University of Rennes and a junior member of the Institut Universitaire de France (IUF), his career trajectory illustrates an evolution from the foundational management of dynamic software architectures to the large-scale analysis of invasive web tracking mechanisms.¹ This dossier provides an exhaustive exploration of his academic contributions, institutional leadership, and scientific vision, structured to serve as both a definitive scholarly record and a narrative expansion of a professional presentation of his research interests.

Academic Identity and Professional Trajectory

The academic journey of Walter Rudametkin is defined by a commitment to international collaboration and a multidisciplinary approach to computer science. His foundational training was established in Mexico at the Universidad Autónoma de Baja California, where he earned a Bachelor’s degree in Computer Sciences, before transitioning to France through the prestigious "Programa de Ingenieros en Francia".¹ This international mobility led to dual degrees from the Ecole Nationale Supérieure d’Informatique et de Mathématiques Appliquées de Grenoble (ENSIMAG) and a Master’s from Université Joseph Fourier.¹

The professional evolution of Walter Rudametkin can be categorized into four distinct institutional phases, each marked by a specific focus in software systems research.

Career Phase	Institution	Primary Research Domain	Key Institutional Roles
Doctoral Research (2007–2013)	Université Grenoble Alpes / LIG / Bull SAS	Dynamic software components and middleware consistency	CIFRE PhD Candidate, R&D Engineer (JOnAS)
Postdoctoral Specialization (2013–2014)	Inria Rennes / Diverse Team	Software diversity and initial privacy exploration	Postdoctoral Researcher

Associate Professorship (2014–2022)	University of Lille / Spirals Team / CRIStAL	Web privacy, AmlUnique, and self-adaptive systems	Associate Professor (Polytech Lille), HDR recipient
Full Professorship (2022–Present)	University of Rennes / IRISA / Inria / IUF	Hardware fingerprinting and software resilience	Full Professor (ESIR), IUF Junior Member

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His doctoral work was notably supported by a CIFRE industrial agreement with Bull SAS, where he contributed to the OW2-JOnAS open-source application server.⁴ This industry-academic hybrid background provided him with a unique perspective on the practical challenges of maintaining high-availability software systems, which later informed his theoretical work on runtime architectural dynamism.⁴

In 2021, he obtained his Habilitation à Diriger des Recherches (HDR) from the University of Lille, a milestone that signaled his readiness to lead large-scale research initiatives.¹ His subsequent election as a junior member of the IUF in 2022 represents one of the highest honors in French academia, acknowledging his exceptional scientific output and the strategic importance of his research into web privacy and self-protecting systems.²

Core Research Interest Pillar I: Dynamic Software Systems and Runtime Reconfiguration

The earliest and most enduring research interest of Walter Rudametkin centers on the inherent dynamism of software architectures. Modern applications are rarely static; they must adapt to changing execution environments, hardware availability, and user requirements without service interruption.¹ This area of research, often termed "runtime architectural dynamism," explores the mechanisms and formalisms required to safely reconfigure a system's structure during execution.¹

Consistency in Component-Based Architectures

The cornerstone of this research pillar is the "Robusta" approach, developed during his Ph.D. research.⁴ Robusta addresses the critical problem of maintaining application consistency despite dynamic changes. When a component-based system is reconfigured—for example, when a software component is replaced by a newer version or a different variant—there is a risk of corrupting the system's state or violating architectural constraints.⁴

The research identified several necessary conditions for safe dynamism:

- **Quiescence and Decoupling:** Ensuring that components are in a "passive" state before reconfiguration to avoid intercepting active requests or leaving the system in an

inconsistent state.⁵

- **Explicit Declarations of Dynamism:** Utilizing architectural models to specify which components can be changed and under what conditions, providing architects with fine-grained control over runtime behavior.⁴
- **Verification Tools:** Developing software analysis tools that can monitor dynamic behavior and verify that reconfigurations do not lead to undesirable or erroneous states.⁴

Component Model	Application Focus	Key Research Contributions
OSGi Service Platform	Dynamic services and modularity	Management of real-time Java applications and service life cycles
Fractal Component Model	Hierarchical architectures	Dynamic instrumentation for tracing and runtime analysis
iPOJO	Service-oriented components	Handling dynamic reconfigurations and component decoupling
JOOnAS	Enterprise middleware	Implementation of OSGi-based JavaEE servers for industry environments

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This work has significant implications for high-availability systems where "stopping" the software for updates is not an option. It extends into pervasive computing environments, such as smart homes and museum exhibits, where the discovery of new devices and services occurs continuously.⁵ For instance, the "NFCMuseum" project demonstrated how dynamic middleware can facilitate a seamless, context-aware experience by reconfiguring its internal services based on the user's proximity to specific artifacts.⁵

Core Research Interest Pillar II: Web Privacy and Stateless Tracking

Transitioning from the structural dynamism of components, Walter Rudametkin applied these principles to the domain of cybersecurity, specifically focusing on the "privacy paradox" of modern web browsers. While browsers add features to improve user experience, these

features often provide "stateless" identification signals known as browser fingerprints.⁹ Unlike traditional cookies, which can be deleted, a fingerprint is a collection of hardware and software attributes that can uniquely identify a user across different sessions and websites.⁹

The AmlUnique Initiative and Empirical Privacy Analysis

The AmlUnique project (amiunique.org) stands as one of the most influential contributions to the field of web privacy.¹⁴ Co-founded and led by Rudametkin, the project serves a dual purpose: it educates the public by allowing users to test their own browser's uniqueness, and it compiles a massive, crowd-sourced dataset for scientific research.¹⁴

The analysis of this data has revealed deep insights into the tracking ecosystem. Utilizing Shannon's entropy as a measure of identifiability, the research quantifies how much "information" a specific browser attribute (e.g., the canvas rendering or the list of installed fonts) contributes to a unique profile.²⁰

Fingerprinting Category	Specific Attributes Analyzed	Privacy Implications
Software Configuration	User agent string, OS version, language, time zone	Basic categorization; high stability over time
Browser Features	Plugin list, cookie settings, DNT header	Identification of specific browser versions and extensions
Graphics and Rendering	Canvas fingerprinting, WebGL characteristics	Hardware-linked signatures; very high entropy
Hardware Artifacts	Screen resolution, CPU cores, GPU execution units	Physical device signatures that persist across OS reinstalls

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One of the most significant findings from this research, published in the award-winning "Beauty and the Beast" paper, was that approximately 90% of desktop browsers and 81% of mobile browsers are unique within a large-scale population.¹⁰ This uniqueness allows advertisers and malicious actors to track users without their consent, circumventing privacy regulations like the GDPR.⁹

The Evolution and Resilience of Fingerprints

The research has moved beyond static uniqueness to explore the "evolution" of fingerprints over time. Through the development of "FP-Stalker," Rudametkin and his colleagues demonstrated that even when a user updates their browser or changes minor settings, the core of their fingerprint remains sufficiently stable to allow for tracking over periods exceeding 100 days.¹⁰ This longitudinal tracking capability presents a severe challenge to existing privacy defenses.

Furthermore, his work on "FP-Scanner" revealed a critical irony: many privacy-enhancing extensions actually make users *more* unique.¹⁰ By attempting to block certain trackers or forge browser attributes, these extensions create "anomalous" fingerprints that do not exist in standard browser installations, effectively flagging the user as a "privacy-conscious individual" and providing a clear signal for identification.¹⁰

Core Research Interest Pillar III: Self-Adaptive and Self-Protecting Software

As a key member of the Spirals research team (a joint initiative between Inria and the University of Lille), Walter Rudametkin contributes to a vision of "self-adaptive software services".⁵ This research aims to automate the management of complex, distributed software systems by introducing autonomous control loops that handle self-optimization, self-healing, and self-protection.²⁸

Architectural Elements for Self-Adaptation

The foundational elements of this pillar involve modeling the variability of software systems. Rudametkin utilizes techniques such as Software Product Lines (SPL) and feature modeling to capture the commonalities and variabilities across different versions of a software product.²⁸ This allows a system to select the most appropriate configuration at runtime based on the current context.⁵

Adaptive Property	Research Objective	Technical Approach
Self-Optimization	Improve performance and resource efficiency	Monitoring KPIs "in the wild" and collaborative decision-making
Self-Protection	Defend against tracking and vulnerabilities	Automated generation of shims and browser reconfiguration
Self-Healing	Maintain availability despite	Dynamic component

	failures	replacement and consistency management
Self-Configuration	Automate setup in heterogeneous environments	Multi-cloud microservice configuration and feature modeling

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In the context of web privacy, "self-protection" involves creating browsers or operating systems that can automatically detect when they are being fingerprinted and respond by reconfiguring their parameters.⁵ This might include generating "automated shims" that replace harmful tracking code with benign placeholders, ensuring the website continues to function while protecting the user's identity.²⁸

The research also explores "debloating," which is the automated removal of unnecessary code and dependencies in modern web and mobile applications.²⁸ By reducing the size of the application, developers can minimize the attack surface, as every external library (like those in NPM or Gradle) represents a potential source of vulnerabilities or malicious code.²⁸

Core Research Interest Pillar IV: Hardware-Level Identification and Side-Channels

A more recent and highly impactful research interest of Walter Rudametkin lies at the intersection of web APIs and hardware micro-architectures. This work investigates how high-performance web features, such as those provided by WebGL, WebGPU, and WebXR, expose subtle hardware characteristics that can be used for identification.²³

DrawnApart: Exploiting GPU Manufacturing Variations

The "DrawnApart" project represents a paradigm shift in fingerprinting research. Unlike previous techniques that relied on software configuration, DrawnApart identifies a device based on minute manufacturing variations in its GPU stack.²³ The research demonstrates that nominally identical GPUs have slight differences in the execution speed of their individual processing units.²⁴

By executing carefully crafted rendering tasks via unprivileged JavaScript, the DrawnApart algorithm can measure these speed variations and generate a unique hardware signature.²³ This signature is remarkably robust and can tell apart computers with identical hardware and software configurations—something previously considered nearly impossible for remote trackers.²³

The technical process involves:

- **Execution Unit Stressing:** Creating rendering pipelines that isolate specific parts of the

GPU.²⁴

- **High-Resolution Timing:** Using unprivileged browser APIs to measure the precise latency of these tasks.²³
- **Deep Learning Classification:** Implementing a deep-learning-based pipeline that moves away from traditional machine learning to better handle the complexities of realistic, "open-world" scenarios.²³

Feature	Description	Impact on Privacy
Target Hardware	Graphics Processing Units (GPUs)	Identifies manufacturing-level differences in silicon
Access Method	Unprivileged JavaScript (WebGL)	Can be run by any website without special permissions
Precision	Can distinguish "identical" devices	Nullifies defenses based on software homogeneity
Tracking Gain	Up to 67% boost in median tracking duration	Dramatically extends the lifespan of a browser fingerprint

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This research highlights a growing threat where the "fingerprinting surface" of a device is no longer limited to the browser but extends deep into the hardware.²² It necessitates new types of defenses that can obscure micro-architectural signatures without sacrificing the performance benefits of modern hardware acceleration.²²

Societal Impact, Demographics, and the Future of the Web

The research of Walter Rudametkin extends beyond technical metrics to explore the societal and demographic implications of digital tracking. Recent studies conducted by his team have shown that fingerprinting risks are not uniform; they vary significantly according to user demographics.³⁵

Demographic Privacy and Data Ethics

By analyzing a first-of-its-kind dataset containing both browser attributes and user-provided demographics (from 8,400 participants), the research identified that lower-income users and

older adults are often at higher risk of being uniquely fingerprinted.³⁵ This may be due to the use of older hardware or less standard configurations that make their devices stand out in a crowd of more modern, homogeneous hardware.³⁵ Furthermore, the research demonstrated a disturbing new risk: demographic information itself (such as age, gender, race, and income) can be inferred from the very browser attributes used for fingerprinting.³⁵ This allows trackers to not only follow a user across the web but also to build detailed psychological and socio-economic profiles without explicit data collection.²¹

The obsolescence of the User Agent

In a 2024 study, Rudametkin and his collaborators investigated the relevance of the User Agent (UA) string, a historical staple of web browsing used to tailor content to specific browsers.³¹ By comparing website behavior when accessed with standard versus "empty" UA headers, the research found that 100% of websites remained functionally identical before JavaScript execution.³¹ After JavaScript execution, only a small fraction of sites changed, mostly due to advertising or bot detection scripts.³¹ This findings suggest that the UA string is increasingly obsolete for its original purpose but remains a significant source of privacy leakage, providing further evidence for the need to move toward more privacy-preserving web standards.²²

Teaching and Institutional Leadership

Beyond his research, Walter Rudametkin is deeply involved in the academic community through teaching and leadership roles. At Polytech Lille and ESIR (University of Rennes), he teaches core computer science subjects, including Object-Oriented Programming (OOP), Software Architecture, and Databases.⁵ He has a passion for hands-on, project-based learning, which he views as essential for training the next generation of software engineers.⁴ His service to the scientific community is extensive:

- **Program Committees:** He serves as a chair and member of program committees for major conferences in software engineering and security (e.g., ICSE, SEAMS, Middleware).¹⁰
- **Awards and Recognition:** He was awarded the CNIL-Inria "protection of privacy" prize in 2018 for his work on browser fingerprinting.⁹
- **IUF Membership:** As a junior member of the IUF, he represents the scientific excellence of French research in information technology.²
- **Mentorship:** He has supervised numerous PhD candidates and postdoctoral researchers, fostering a new generation of experts in web privacy and software diversity.¹

University	Role	Teaching Concentrations	Research Group
University of Lille (Polytech)	Associate Professor	OOP, Architecture, Databases	Spirals Team (CRIS ² AL / Inria)

University of Rennes (ESIR)	Full Professor	Advanced Software Engineering	Diverse Team (IRISA / Inria)
Inria Rennes	Postdoctoral / Researcher	Software Diversity, Privacy	Diverse (formerly Triskell)
Université Grenoble Alpes	PhD Candidate / Instructor	Component Systems	Adèle Research Team

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Comprehensive Summary of Research Philosophy

The scientific worldview of Walter Rudametkin can be summarized as a search for "predictable dynamism." Whether he is architecting a middleware platform that must update itself during a live transaction or designing a browser extension that must shield a user from invisible trackers, the goal is to manage change in a way that preserves both functionality and security.¹

His work has demonstrated that:

1. **Software Diversity is a Defense:** By automatically creating and rotating software variants, we can prevent attackers from relying on a fixed set of target characteristics.⁵
2. **Privacy is an Engineering Challenge:** Effective privacy protection requires a deep understanding of the software stack, from the high-level application code down to the physical characteristics of the silicon.¹³
3. **Transparency Requires Tools:** Large-scale empirical projects like AmlUnique are necessary to provide the data needed for informed policy decisions and technical innovations.⁹
4. **Self-Adaptation is Essential:** As the web grows in complexity, the only way to ensure privacy and efficiency is to build systems that can manage themselves.²⁸

Through his multi-faceted role as a researcher, educator, and scientific leader, Walter Rudametkin continues to push the boundaries of what is possible in the creation of a more secure, private, and resilient digital future. His work bridges the gap between the theoretical foundations of software engineering and the urgent, real-world needs of millions of web users.

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